Design and Application of a GQIM-Based Software Measurement Plan

SWE 6763 Assignment 2
Group 2

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Abstract

This document presents a Goal–Question–Indicator–Metric (GQIM) based software measurement plan for a Smart Banking Platform. The primary objective is to ensure data integrity and security across all banking operations, maintaining customer trust, reliable transactions, and regulatory compliance. Guided by the GQIM framework discussed in SWE 6783 under Dr. Md Shazibul Islam Shamim, the team defined a business goal, derived measurement sub-goals, identified key entities and attributes, formalized indicators and measures, and developed an implementation plan with quantitative thresholds and corrective actions. The plan supports continuous improvement through structured monitoring and clear operational targets.

Step 1: Business Goal (Identify Business Goals)

Business Goal: Ensure data integrity and security across all banking operations to protect user assets and maintain regulatory compliance.

This goal reflects the organization's emphasis on customer trust, operational reliability, and alignment with global data-protection and financial transparency requirements. It provides a foundation for the specific measurement goals that follow.

Step 2: Key Knowledge Areas (Identify What You Want to Know or Learn)

The team focused on assessing how effectively the platform safeguards information and maintains operational accuracy. Insights were developed through analysis of current processes, system logs, and compliance requirements:

- Accuracy and consistency of financial transactions.
- Authenticity and tamper-resistance of stored data.
- Strength of authentication, encryption, and access control mechanisms.
- Reliability of system up-time and recovery procedures.
- Adherence to regulatory and audit standards.

This understanding guided the identification of sub-goals and measurable entities.

Step 3: Identified Sub-goals (Identify Sub-goals)

Based on these findings, the team established five measurement sub goals representing the platform's core performance dimensions:

1. **Transaction Accuracy** – Confirm that all financial transactions are processed and recorded correctly.
2. **Data Integrity** – Maintain accurate and traceable information across systems and backups.
3. **Access Security** – Restrict access to authorized users and enforce strong authentication.
4. **System Availability** – Maintain consistent up-time and fast recovery from disruptions.
5. **Compliance and Audit Readiness** – Demonstrate adherence to data-protection and financial-reporting standards.

Step 4: Entities and Attributes (Identify Entities and Attributes Related to Sub-goals)

For each sub-goal, the team identified entities and measurable attributes derived from system logs, monitoring dashboards, and audit reports.

Entity	Key Attributes to Measure
Transaction record	Correctness, reconciliation status, timestamp, balance consistency.
Database change event	Authorization flag, log completeness, trace ID, data checksum.
User access event	Login success/failure, MFA usage, encryption status, privilege level.
Service instance	Up-time percentage, downtime duration, incident frequency, recovery time.
Compliance control	Control test status, audit finding severity, remediation time.

Step 5: Formalized Measurement Goals (Formalize Measurement Goals)

Each measurement goal was documented with its object, purpose, perspective, and environmental constraints.

G1 – Transaction Accuracy

Object: Transaction record.
Purpose: Measure the precision of transaction postings and account reconciliations.
Perspective: Banking operations management.
Environment & Constraints: Online and mobile platforms; dependent on network reliability and third-party payment gateways.

G2 – Data Integrity

Object: Database change events and audit logs.
Purpose: Detect unauthorized data modifications and preserve historical traceability.
Perspective: Security administration.
Environment & Constraints: Centralized database; governed by retention and access-control policies.

G3 – Access Security

Object: User access events.
Purpose: Evaluate the enforcement of authentication, authorization, and encryption standards.
Perspective: Information security management.
Environment & Constraints: Multi-factor authentication (MFA) and least-privilege models; managed under IAM policies.

G4 – System Availability

Object: Service uptime records and incident logs.
Purpose: Monitor reliability and responsiveness of the platform under real usage.

Perspective: DevOps and infrastructure teams.
Environment & Constraints: Cloud-hosted infrastructure; limited by maintenance schedules and redundancy budgets.

G5 – Compliance and Audit Readiness

Object: Compliance controls and audit findings.
Purpose: Evaluate adherence to internal and external regulatory requirements.
Perspective: Compliance and risk management.
Environment & Constraints: Data-protection and financial-reporting laws with defined audit timelines.

Step 6: Indicators (Identify Quantifiable Questions and Indicators Used to Achieve Measurement Goals)

Indicators were defined to measure progress toward each goal, leveraging existing data sources and monitoring tools.

- **Accuracy Ratio** – Ratio of verified transactions to total transactions.
- **Integrity Score** – Frequency of unauthorized data changes and audit-trail completeness.
- **Security Index** – Compliance with MFA and least-privilege access standards.
- **Availability Rate** – Percentage uptime and mean time to recover (MTTR).
- **Compliance Score** – Percentage of controls passed during audits and severity of nonconformities.

Step 7: Data Elements (Identify Data Elements Collected to Construct Indicators)

Data elements were identified from operational logs, monitoring tools, and compliance repositories.

Indicator	Primary Data Elements
Accuracy Ratio	Transaction IDs, reconciliation flags, timestamps.
Integrity Score	Log entry IDs, change types, authorization flags, audit log completeness fields.
Security Index	Login attempts, MFA status, user-role assignments, encryption state.
Availability Rate	Uptime minutes, downtime minutes, incident identifiers, recovery timestamps.
Compliance Score	Control identifiers, test results, audit findings, remediation completion dates.

Step 8: Measures and Thresholds (Define Measures and Make Them Operational)

The team established formal quantitative definitions and thresholds for each measure.

Transaction Accuracy

$$\boxed{\text{Transaction Accuracy} = \frac{\text{Verified Transactions}}{\text{Total Transactions}} \times 100\%}$$

Percentage of successfully verified transactions. **Interpretation:** $\geq 99\%$ excellent; 95–98% acceptable; $< 95\%$ critical—requires investigation.

Data Consistency

$$\boxed{\text{Data Consistency} = \frac{\text{Accounts with Zero Discrepancy}}{\text{Total Accounts}} \times 100\%}$$

Ratio of accounts without discrepancies between balance and transaction history. **Interpretation:** $\geq 98\%$ strong integrity; 95–97% minor inconsistencies; $< 95\%$ major issues.

Access Security Compliance Ratio (ASCR)

$$\boxed{\text{ASCR} = \frac{\text{Principals Meeting All Access Controls}}{\text{Total Active Principals}} \times 100\%}$$

Proportion of users and systems meeting MFA, least-privilege, and access-policy requirements. **Targets:** $\geq 98\%$ optimal; $< 95\%$ requires immediate remediation.

MFA Coverage

Definition: Two measures—(a) Users with effective MFA; (b) Service accounts using strong non-password authentication. **Targets:** Users $\geq 98\%$; Administrators $= 100\%$; Service Accounts $= 100\%$.

Security Incident Metrics

Data-leakage incidents: Zero confirmed incidents. **Anomaly detection time:** ≤ 24 hours. **Mean Time to Recovery (MTTR):** ≤ 4 hours for critical services.

Availability and Compliance

Uptime: Maintain monthly average $\geq 99.9\%$. **Audit performance:** All high-severity findings remediated within established deadlines.

Step 9: Actions (Identify Actions Taken to Implement the Measures)

This section focuses on transforming measurement results into meaningful improvements. The team uses a structured, three-phase approach—**Analysis**, **Diagnosis**, and **Action**—to interpret data, identify underlying causes of deviations, and implement targeted corrective measures. This continuous feedback loop ensures that insights gained from monitoring directly enhance the reliability, security, and performance of the Smart Banking Platform.

Analysis	**Diagnosis**	**Action**
Review transaction accuracy, data integrity, and system availability to detect deviations from targets.	Identify causes such as reconciliation logic errors, incomplete audit logs, or server latency.	Correct data issues, refine reconciliation scripts, and deploy patches within one week.
Analyze ASCR and MFA coverage reports for compliance gaps.	Trace issues to inactive policies, role drift, or unlinked identity sources.	Re-enforce MFA policies, reassign privileges, and verify compliance in the next cycle.
Evaluate incident and MTTR trends from dashboards.	Determine whether delays stem from resource shortages, outdated run-books, or escalation issues.	Update incident-response plans, automate recovery, and conduct postmortems within 14 days.
Compare compliance scores across audit cycles.	Detect recurring control gaps or delayed remediation.	Implement corrective actions, document updates, and re-test controls before next audit.

This structured approach ensures that measurement data directly leads to actionable improvements and continuous system refinement.

Step 10: Implementation Plan (Prepare a Plan for Implementing the Measures)

This section presents the implementation plan for executing the measurement framework within the Smart Banking Platform. It describes the workflow, responsibilities, and timing required to operationalize the defined metrics through consistent data collection, analysis, reporting, and feedback. The plan ensures that measurement activities are sustainable, traceable, and integrated into daily operations for ongoing performance improvement.

Phase	Planned Activities	Ownership	Cadence
Planning	Define and validate metrics, thresholds, data sources, and tools; establish data-collection methods and reporting scope.	Measurement Team, DevOps Lead, Compliance Officer	Once (project initiation)
Data Collection	Automate log and transaction data ingestion; ensure completeness, accuracy, and timeliness.	Security Engineer, DevOps Engineer	Continuous / Weekly roll-ups
Storage and Management	Maintain secure data storage with version control, access management, and retention policies.	Database Administrator, Compliance Analyst	Continuous / Monthly integrity checks
Analysis and Reporting	Compute indicators, generate dashboards, and deliver performance summaries.	Measurement Analyst, Security Analyst	Weekly analysis / Monthly reporting
Review and Feedback	Evaluate measurement results, identify deviations from goals, and prioritize improvement actions.	CISO, Product Manager, Compliance Officer	Monthly tactical / Quarterly strategic reviews
Improvement and Evolution	Implement corrective actions, refine thresholds, and update data-collection processes as systems mature.	DevOps, Security, and Compliance Teams	Quarterly / As required

Expected Outcomes and Success Metrics: Each phase of the implementation plan contributes measurable results and performance targets. Expected outcomes include a validated metrics catalog, automated data pipelines, secure storage repositories, and timely analytics that inform management decisions. Success will be evaluated using the following key indicators:

Success Metrics:

- Transaction Accuracy $\geq 99\%$; Data Consistency $\geq 98\%$.
- Zero data-leakage incidents; detection $\leq$ 24h; recovery $\leq$ 4h.
- ASCR $\geq 98\%$; MFA compliance 100%.
- System uptime $\geq 99.9\%$ monthly.
- All compliance findings remediated within policy deadlines.

These outcomes demonstrate successful execution of the GQIM measurement plan and continuous improvement of the Smart Banking Platform's operational performance.

Summary

The Smart Banking Platform measurement plan demonstrates how goal-based metrics transform organizational objectives into measurable indicators. Through quantitative thresholds, trigger-based actions, and defined review cycles, the plan establishes a continuous improvement framework supporting data integrity, security, system reliability, and compliance. This structure enables consistent monitoring, timely remediation, and stronger accountability across operational teams.

References

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